

# Gaurav Singh

Full-Stack & AI Software Engineer | Distributed Systems & Machine Learning

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## TECHNICAL SKILLS

**Languages:** Python, JavaScript (ES6+), TypeScript, C++, SQL, HTML5/CSS3, Bash

**Frontend & UI:** React, Next.js, Electron, Three.js, GSAP, HTML5 Canvas, WebGL, Streamlit

**Backend & APIs:** Node.js, FastAPI, RESTful APIs, Microservices, OpenRouter API, WebRTC, Web Speech API

**AI & Machine Learning:** TensorFlow, TensorFlow.js, OpenCV, Deep Learning (CNNs), Grad-CAM, Pandas, Plotly.js

**Databases & Developer Tools:** PostgreSQL, Redis, IndexedDB, Docker, Git/GitHub, Linux (Bash), Postman, CI/CD

**Operating Systems:** Linux (Arch Linux, Ubuntu, Debian), macOS, Windows

## PROJECTS

### Aiely | Electron, Node.js, OpenRouter API, Web Speech API

- Engineered a cross-platform AI desktop copilot for macOS and Windows with a native floating overlay that remains hidden during screen sharing and recordings.
- Built a real-time context engine combining screen analysis with dual-channel speech recognition (microphone + system audio) to generate instant coding assistance and meeting responses.
- Designed a multi-model failover architecture using OpenRouter with local-only configuration and zero telemetry, ensuring reliable AI inference, enhanced privacy, and uninterrupted operation.

### BrainScan | TensorFlow.js, JavaScript, HTML/CSS, IndexedDB

- Engineered a privacy-first, browser-based AI application for brain tumor classification using TensorFlow.js with WebGL-accelerated client-side inference.
- Built advanced visualization tools including Grad-CAM heatmaps, an interactive before/after comparison slider, and a magnification lens for enhanced diagnostic interpretability.
- Reduced repeat model load times by implementing IndexedDB caching for model weights, enabling fully serverless deployment and zero patient data transmission.

### F1 Strategy Lab | Next.js, TypeScript, FastAPI, Python, Plotly.js

- Architected a full-stack telemetry analysis platform that processes official Formula 1 race data to deliver interactive performance and strategy insights.
- Implemented high-performance data visualizations including driver telemetry comparison, circuit mapping, speed heatmaps, and lap delta analytics using Plotly.js.
- Designed backend APIs and analytics modules for tyre strategy evaluation, pit stop analysis, and recommendation generation using FastAPI and FastF1.

### PhotoBooth | React, Three.js, GSAP, HTML5 Canvas

- Engineered a real-time interactive graphics application using Three.js and GSAP, implementing mathematically interpolated curtain simulations and physics-inspired animation pipelines.
- Designed computational geometry and image-processing algorithms including AABB collision detection, procedural film grain synthesis, texture generation, and canvas-based photo compositing.
- Built a responsive React application integrating WebRTC camera APIs with GPU-accelerated rendering and optimized animation workflows for seamless performance across desktop and mobile devices.

## EXPERIENCE

### Web Developer Intern

Sep 2025 – Oct 2025

SkillCraft Technology

India, Remote

- Developed responsive web applications using modern HTML, CSS, and JavaScript, following industry best practices.
- Completed a series of hands-on development tasks and projects, implementing interactive UI components and responsive layouts.
- Strengthened front-end development, debugging, and problem-solving skills by building and refining real-world web solutions.

## CERTIFICATIONS

**Harvard University:** CS50's Introduction to Programming with Python

**Deloitte Australia (Forage):** Technology Job Simulation

**Postman:** API Fundamentals Student Expert (Canvas Credentials)

**HackerRank:** Python (Basic) Certification

## EDUCATION

### Bachelor of Engineering in Computer Science

2027

Visvesvaraya Technological University

Bangalore, IN

### All India Senior School Certificate Examination (AISSCE)

2022

Army Public School PRTC

Bangalore, IN

### All India Secondary School Examination (AISSE)

2020

Army Public School

India